University of the Philippines Diliman
College of Engineering
Electrical and Electronics Engineering Institute

First-Order Circuits Part 1

1st Semester, A.Y. 2026–2027

Student Outcomes

- Determine the initial conditions and time constants (τ) of RL and RC circuits.
- Formulate and solve first-order linear homogeneous and non-homogeneous differential equations.
- Illustrate/sketch the complete response by distinguishing between transient and steady-state components.

Lecture Outline

- Characteristics of First-Order Circuits
- Intuitive Overview of a Transient Response
- Homogeneous and Non-Homogeneous Differential Equations
- Transient and Steady-State Solutions

First-Order Circuits

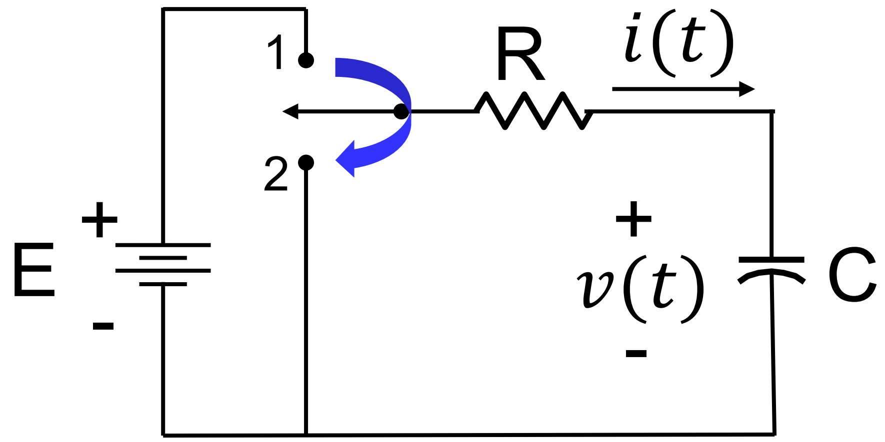
- Any circuit with **exactly one independent energy-storage** element, an arbitrary number of sources, and an arbitrary number of resistors
- Any voltage or current in such a circuit is the solution to a first-order differential equation (DE)
- RL and RC networks are first-order circuits

Note: Energy-storage elements are independent if they cannot be reduced to a single equivalent element via series or parallel combinations.

Characteristics of R, L, and C

Element	VI Relation	DC Steady-state
Resistor	$v_R(t) = Ri_R(t)$	$V = IR$
Inductor	$v_L(t) = L\frac{di_L(t)}{dt}$	$V = 0$; short
Capacitor	$i_C(t) = C\frac{dv_C(t)}{dt}$	$I = 0$; open

Example 1

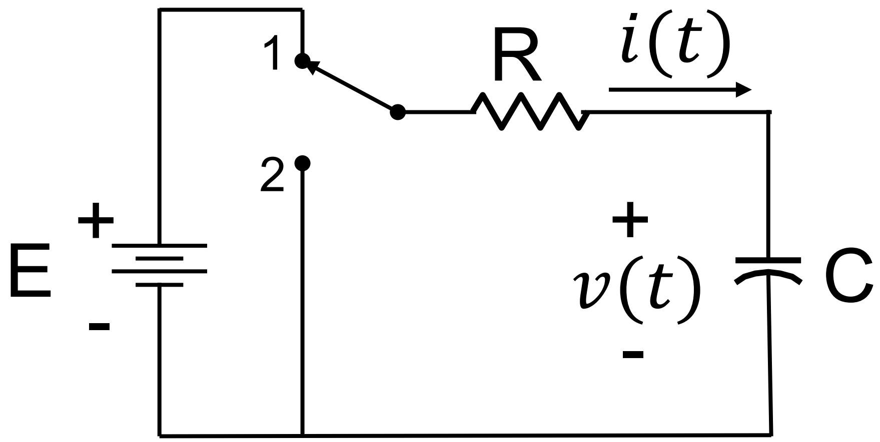


Consider the circuit above.

The switch was in position 1 for a very, very long time. At $t = 0$, the switch is moved to position 2.

Sketch the waveforms for $i(t)$ and $v(t)$ for all time t.

Example 1



Right before the switch changes position:

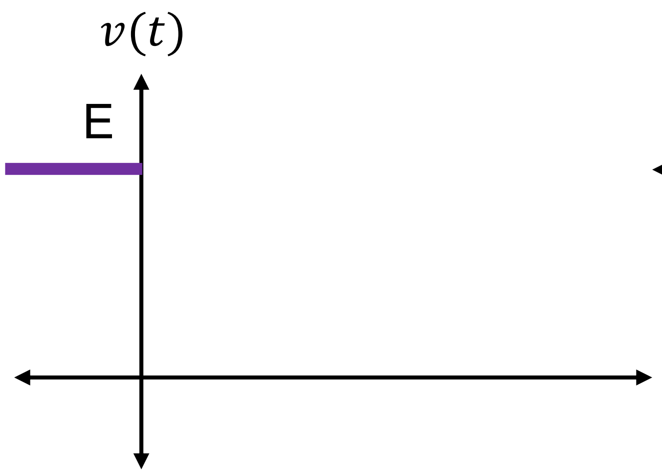
The capacitor voltage and current are in steady-state:

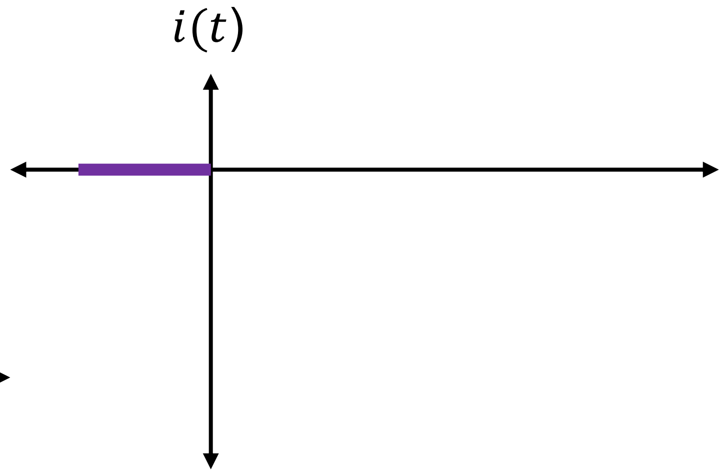
$$i(0^-) = 0$$

$$v(0^-) = E$$

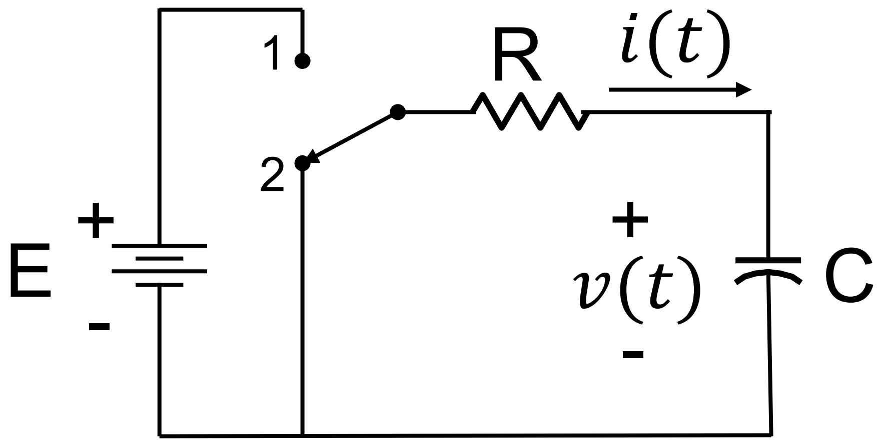
The capacitor is equivalent to an open circuit.

At $t = 0^-$:





Example 1



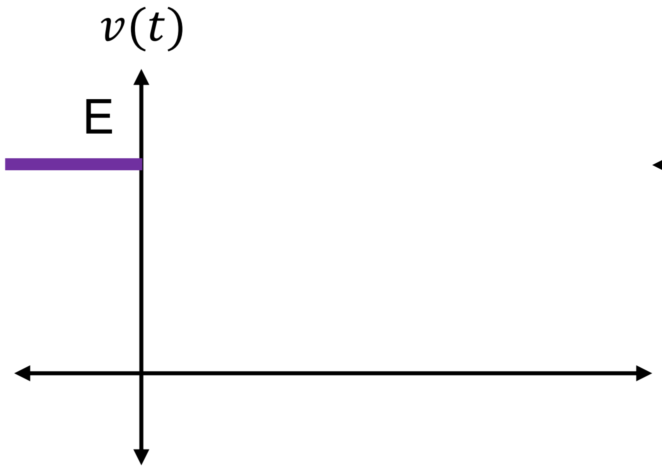
Right after the switch changes position:

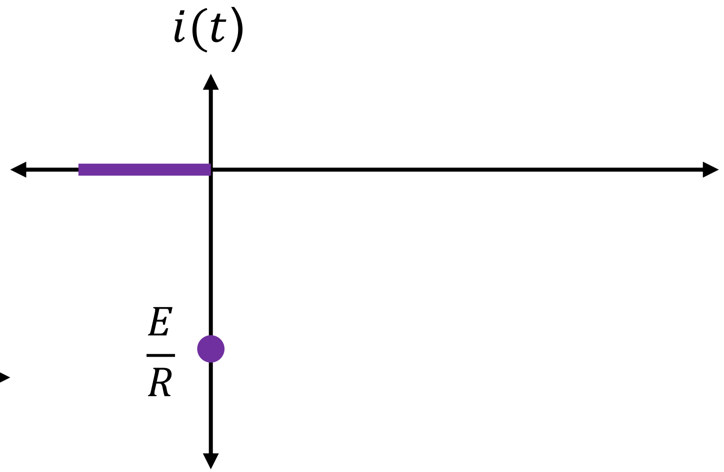
a.) The capacitor voltage remains the same because capacitor voltage cannot change instantaneously

b.) Current starts flowing through the circuit with an initial value $i(0^+) = \frac{E}{R}$

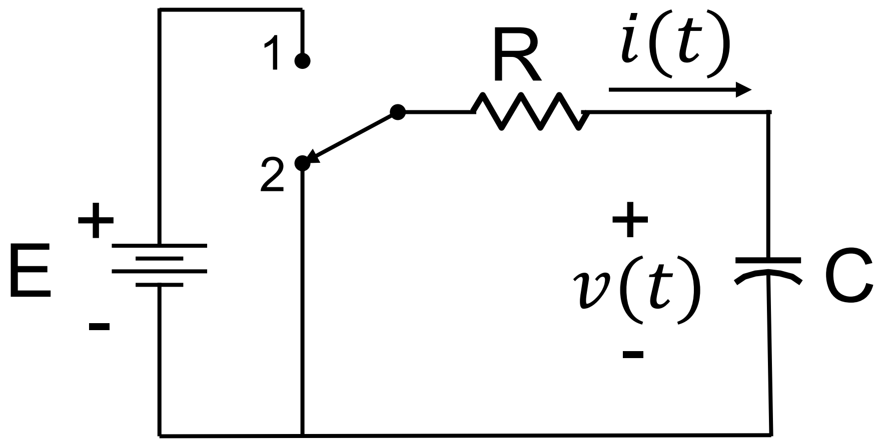
c.) The capacitor is no longer equivalent to an open circuit.

At $t = 0^+$:





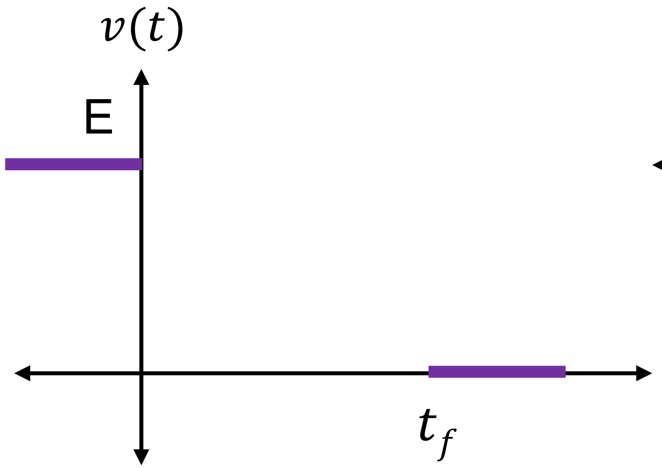
Example 1

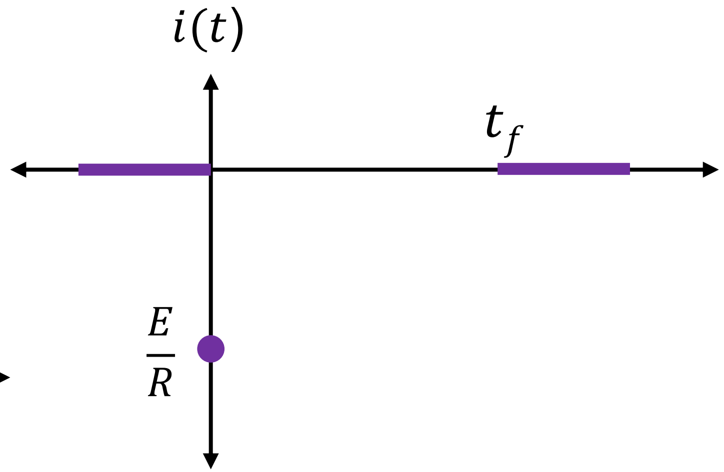


After a "long, long time" t_f:

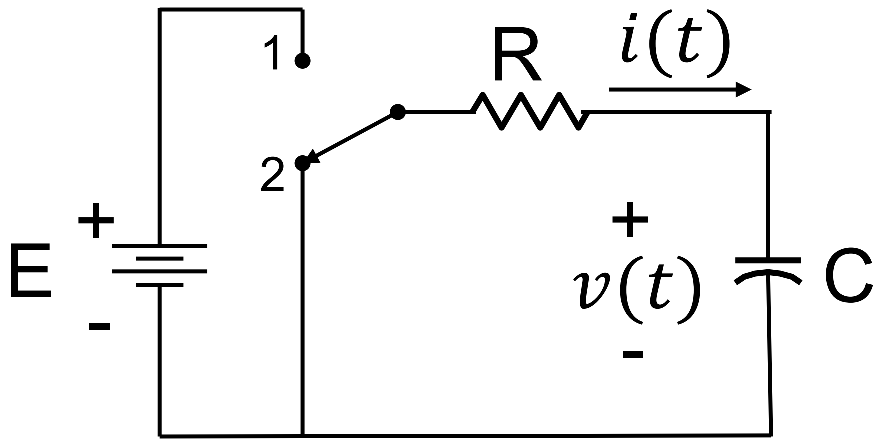
After a time t_f, the energy stored in the capacitor's electric field would be dissipated through the resistor R.

After such time, both voltage $v(t)$ and current $i(t)$ are zero.



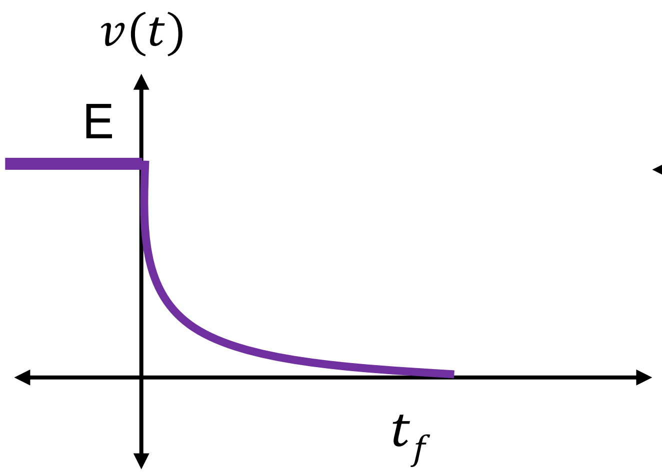


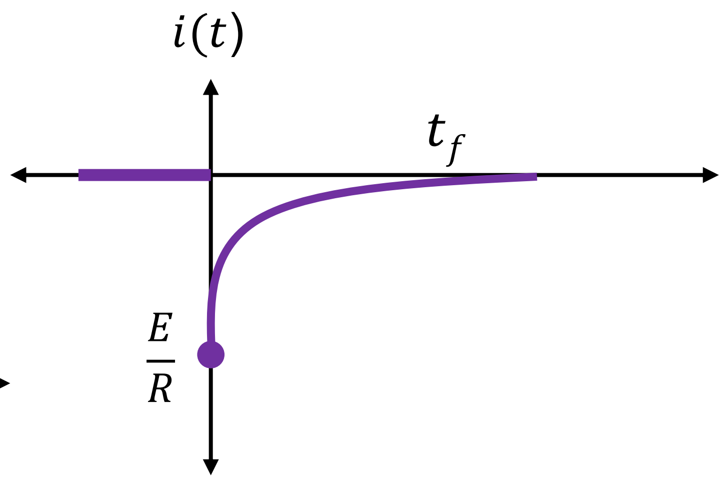
Example 1



Between $t = 0^+$ and $t = t_f$:

The capacitor voltage and current both follow an **exponential response** as they transition from their initial state at $t = 0^+$ to their final state at $t = t_f$.





Differential Equations

General Equation:

$$a_n \frac{d^n x}{dt^n} + a_{n-1} \frac{d^{n-1} x}{dt^{n-1}} + \cdots + a_1 \frac{dx}{dt} + a_0 x = v(t)$$

A **linear ordinary differential equation** describing linear electric circuits has:

- $a_n, a_{n-1}, \dots, a_0$ — constants
- $x(t)$ — dependent variable (current or voltage)
- t — independent variable (time)
- $v(t)$ — voltage or current sources

Differential Equations

Complete Solution:

Its complete solution $x(t)$can be decomposed into the *homogeneous* and *particular* solutions, i.e.,

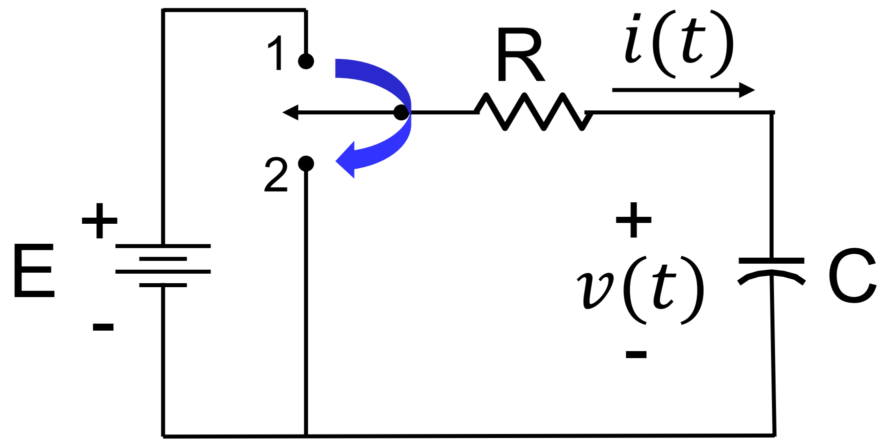
$$x(t) = x_h(t) + x_p(t)$$

Where $x_h(t)$ is any solution to the homogeneous differential equation:

$$a_n \frac{d^n x}{dt^n} + a_{n-1} \frac{d^{n-1} x}{dt^{n-1}} + \cdots + a_1 \frac{dx}{dt} + a_0 x = 0$$

And $x_p(t)$ is any solution to the original non-homogeneous differential equation.

From Example 1



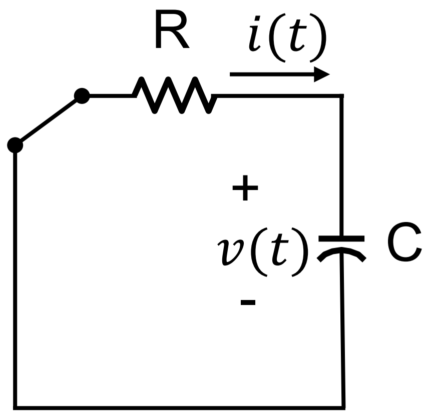
The switch was in position 1 for a very, very long time. At $t = 0$, the switch is moved to position 2.

Determine an expression for the voltage $v(t)$ for $t > 0$.

From Example 1

For $t < 0$, the voltage source is connected. We assume that the capacitor voltage has reached steady-state ("a very, very long time") by $t = 0^-$.

For $t > 0$, our circuit will look like:



The differential equation describing our circuit for $t > 0$ can be obtained via KCL at the top node:

$$C\frac{dv(t)}{dt} + \frac{v(t)}{R} = 0$$

From Example 1

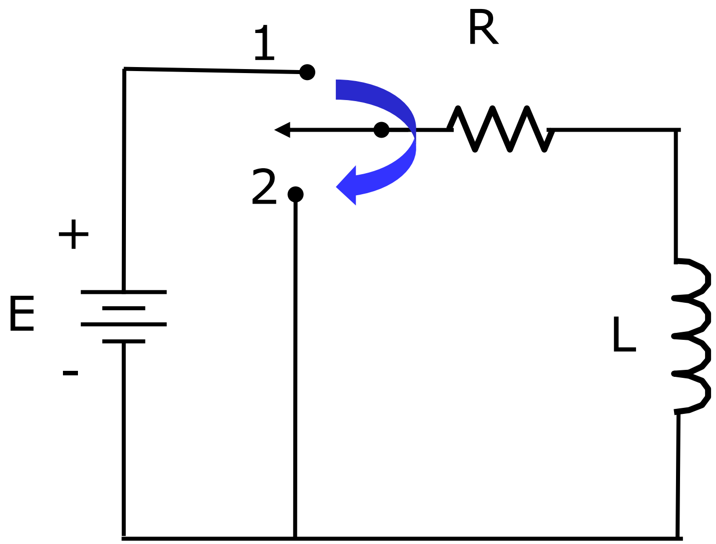
By rearranging the terms, we multiply both sides by R to obtain the standard form:

$$RC\frac{dv(t)}{dt} + v(t) = 0$$

- This is classified as a **homogeneous differential equation** because the right-hand side is equal to zero (meaning there is no external source driving the system for $t > 0$). Its solution is known as the **homogeneous solution**.

- We also refer to this solution as the **natural response** or **unforced response** since it is dependent on the nature of the circuit itself and not on any external voltage or current source acting on it.

Solving Homogeneous Differential Equations



Consider the network shown. The switch is moved from position 1 to position 2 at time $t = 0$.

Applying KVL:
After switching, the KVL equation is

$$L\frac{di}{dt} + Ri = 0 \qquad (1)$$

Separating Variables:
Rearrange terms to isolate the current i and time t variables:

$$\frac{di}{i} = -\frac{R}{L}dt \qquad (2)$$

Solving Homogeneous Differential Equations

Integrating Both Sides:
Integrating equation (2) with respect to their respective variables gives:

$$\ln i = -\frac{R}{L}t + K \qquad (3)$$

where K is the constant of integration.

Using Logarithm Rules to Solve for i:
We can express the constant K as $\ln k$.

$$\ln i = \ln e^{-\frac{R}{L}t} + \ln k \qquad (4)$$

Applying logarithmic properties ($\ln y + \ln z = \ln y\, z$) yields:

$$\ln i = \ln\left(ke^{-\frac{R}{L}t}\right) \qquad (5)$$

Solving Homogeneous Differential Equations

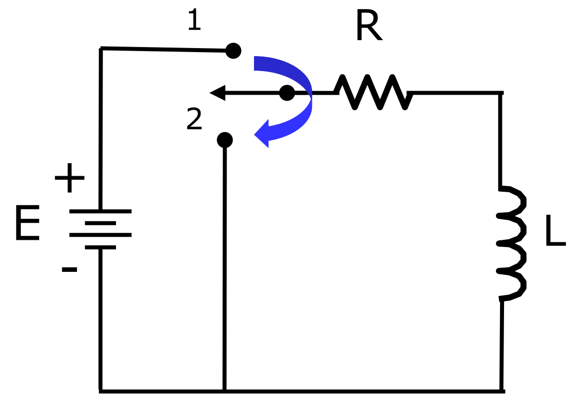
Taking the antilogarithm results in the final time-domain response:

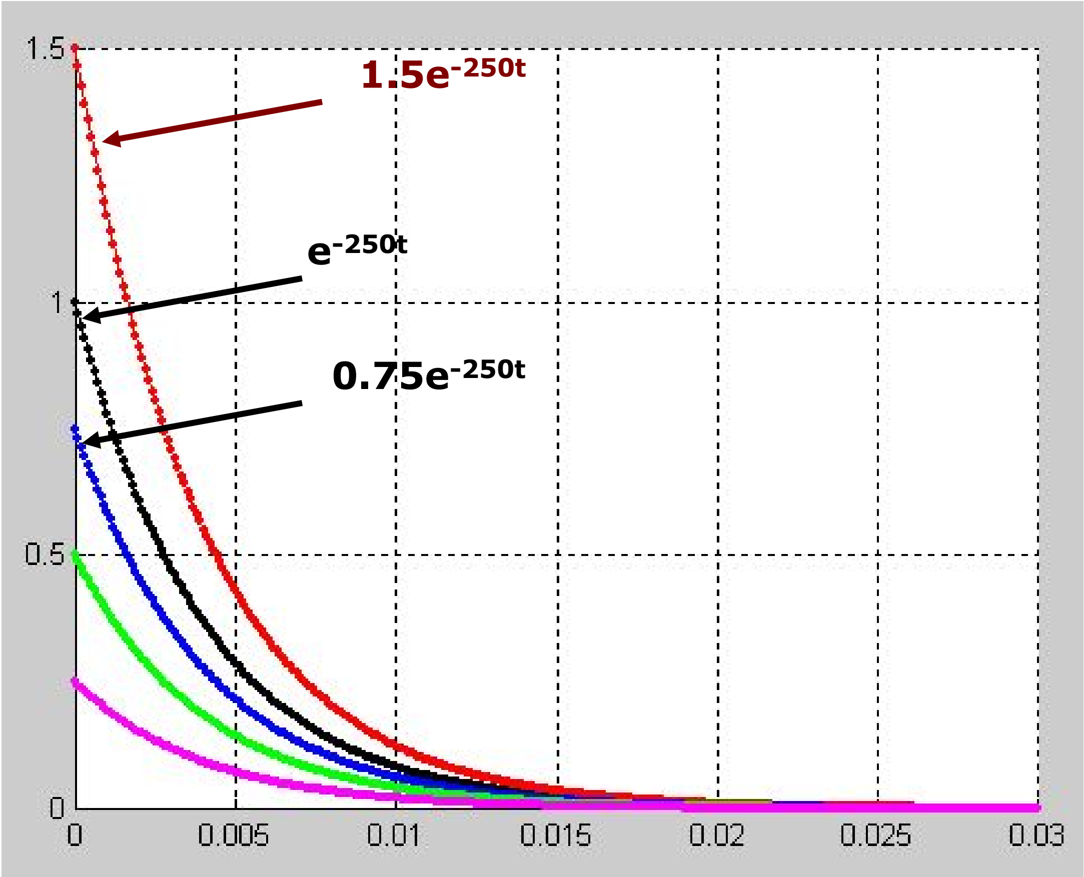
$$i(t) = ke^{-\frac{R}{L}t} \qquad (6)$$

Equation 6 is the form of the natural response of the current $i(t)$, where k can be determined using the initial conditions.

Simply, **the homogeneous solution is an exponential function**.

Solving Homogeneous Differential Equations





Assuming that in the circuit,

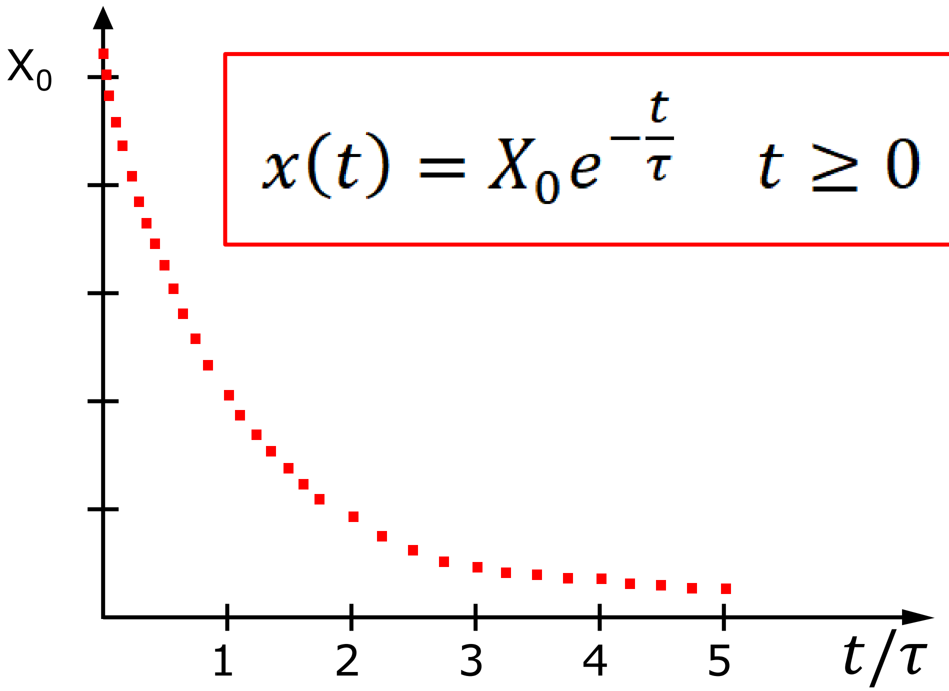
$$R = 1\ \text{k}\Omega, \quad L = 4\ \text{H}$$

then the figure on the right shows the plot of the natural response for different initial conditions of the current $i(t)$.

The Exponential Function

Given the function $x(t) = X_0 e^{-\frac{t}{\tau}}$:

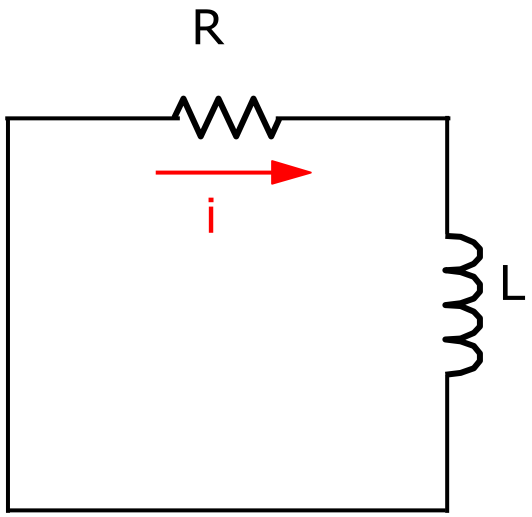
- When $t = 0$, $x(0) = X_0 e^0 = X_0$
- When $t = \tau$, $x(\tau) = X_0 e^{-1} = 0.368 X_0$
- When $t = 2\tau$, $x(2\tau) = X_0 e^{-2} = 0.135 X_0$
- When $t = 3\tau$, $x(3\tau) = X_0 e^{-3} = 0.050 X_0$
- When $t = 4\tau$, $x(4\tau) = X_0 e^{-4} = 0.018 X_0$
- When $t = 5\tau$, $x(5\tau) = X_0 e^{-5} = 0.007 X_0$



$$\boxed{x(t) = X_0 e^{-\frac{t}{\tau}} \quad t \geq 0}$$

Note: After $\boldsymbol{t = 5}\tau$ or after 5 time constants, the function is practically zero.

Example 2: Source-Free RL Network



Consider the circuit shown. Let $i(0) = I_0$.

From KVL, we get:

$$L\frac{di}{dt} + Ri = 0$$

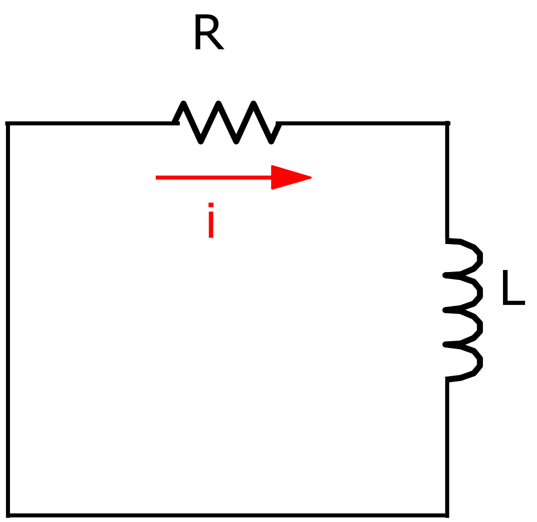
The solution can be found to be:

$$i = Ke^{-\frac{R}{L}t}$$

At $t = 0$, we get:

$$i(0) = I_0 = Ke^0 = K$$

Example 2: Source-Free RL Network



Substitution gives:

$$i(t) = I_0 e^{-\frac{R}{L}t}$$

From Ohm's Law, we get the resistor voltage.

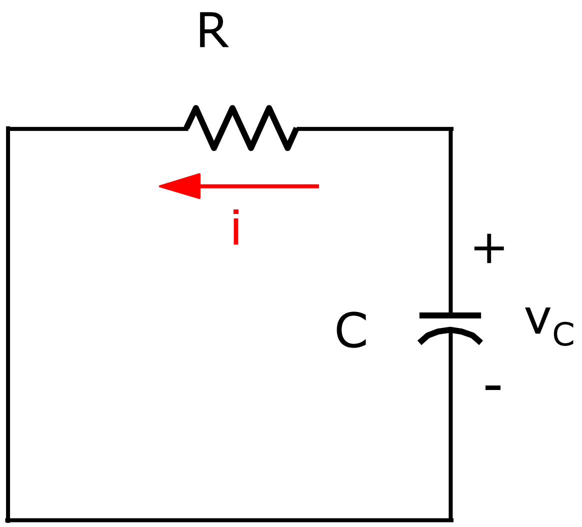
$$v_R = Ri = RI_0 e^{-\frac{R}{L}t}$$

The voltage across the inductor is given by:

$$v_L = L\frac{di}{dt} = -RI_0 e^{-\frac{R}{L}t} = -v_R$$

Note: Every current and voltage in an RL network is a decaying exponential with a time constant of $\boldsymbol{\tau = \frac{L}{R}}$.

Example 3: Source-Free RC Network



Consider the circuit shown. Let $v_C(0) = V_0$.

From KCL, we get for $t \geq 0$:

$$C\frac{dv_C}{dt} + \frac{1}{R}v_C = 0$$

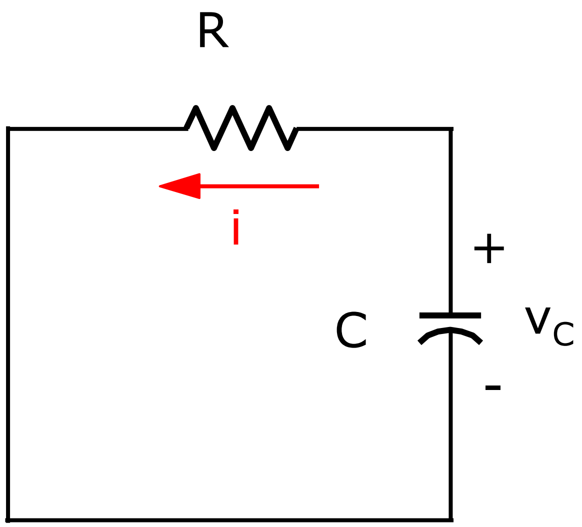
The solution can be shown to be:

$$v_C = Ke^{-\frac{1}{RC}t}$$

At $t = 0$, we get:

$$v_C(0) = V_0 = Ke^0 = K$$

Example 3: Source-Free RC Network



Substitution gives:

$$v_C(t) = V_0 e^{-\frac{1}{RC}t}$$

From Ohm's Law, we get the resistor current.

$$i_R = \frac{v_C}{R} = \frac{V_0}{R} e^{-\frac{1}{RC}t}$$

The current in the capacitor is given by:

$$i_C = C\frac{dv_C}{dt} = -\frac{V_0}{R} e^{-\frac{1}{RC}t} = -i_R$$

Note: Every current and voltage in an RC network is a decaying exponential with a time constant of $\boldsymbol{\tau = RC}$.

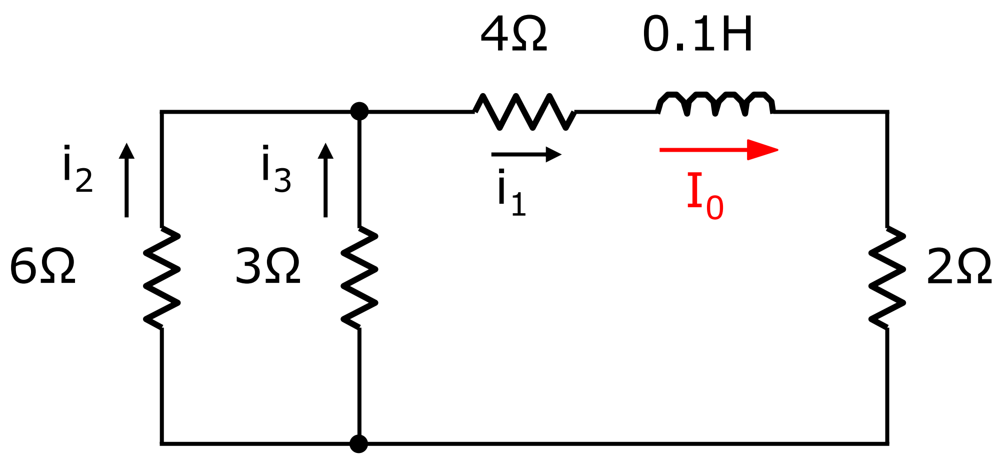
Remarks

1. When R is expressed in ohms, L in Henries and C in Farads, the time constant is in seconds.
2. For practical circuits, the typical values of the parameters are: R in ohms, L in mH, C in μF.
3. Typically, $\tau = \frac{L}{R}$ in msec

   $\tau = RC$ in μsec

Note: For practical circuits, the exponential function will decay to zero in less than 1 second.

Example 4: General RL Circuit



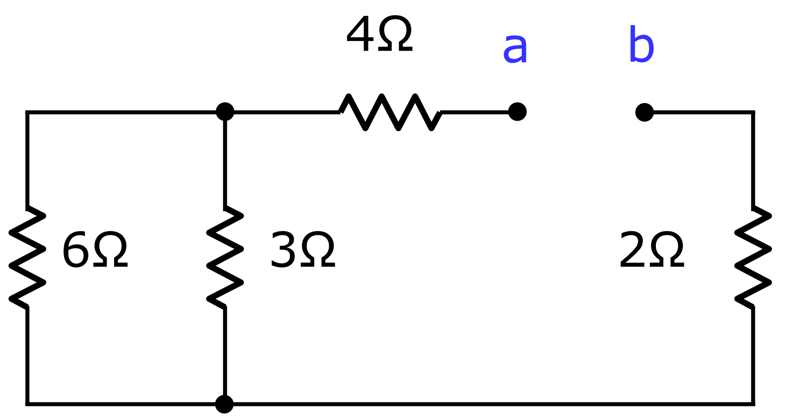
The circuit shown has several resistors but only one inductor. Given

$$i_1(0^+) = I_0 = 2\ \mathrm{A}$$

Find i_1, i_2, and i_3 for $t \geq 0$.

Example 4: General RL Circuit

Solution:



Equivalent Resistance:

First, determine the equivalent resistance seen by the inductor.

$$R_{ab} = 2 + 4 + \frac{6(3)}{6+3} = 8\ \Omega$$

Time Constant (τ):

Using the equivalent resistance $R_{ab} = 8\ \Omega$ and inductance L:

$$\tau = \frac{L}{R_{ab}} = \frac{0.1}{8} = \frac{1}{80} \text{ sec}$$

Example 4: General RL Circuit

Every current in this circuit is described by the exponential form:

$$Ke^{-80t} \quad t \geq 0$$

Solving for K_1:

For current i_1:

$$i_1 = K_1 e^{-80t} \quad t \geq 0$$

Using the initial condition at $t = 0^+$ where $i_1(0^+) = I_0 = 2\text{ A}$:

$$i_1(0^+) = 2 = K_1 e^0 = K_1$$

Thus, $K_1 = 2$.

Example 4: General RL Circuit

Thus, we find the current i_1 to be:

$$\boxed{i_1 = 2e^{-80t}\ \text{A} \quad t \geq 0}$$

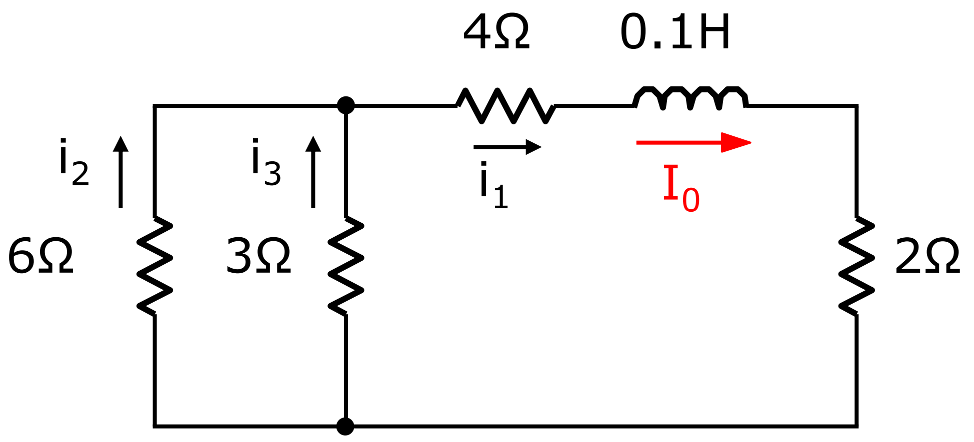
The remaining currents, $i_2$and i_3, can be found using current division. We get:

$$i_2 = \frac{3}{3+6} i_1 = \frac{1}{3} i_1$$

$$\boxed{i_2 = \frac{2}{3} e^{-80t}\ \text{A} \quad t \geq 0}$$

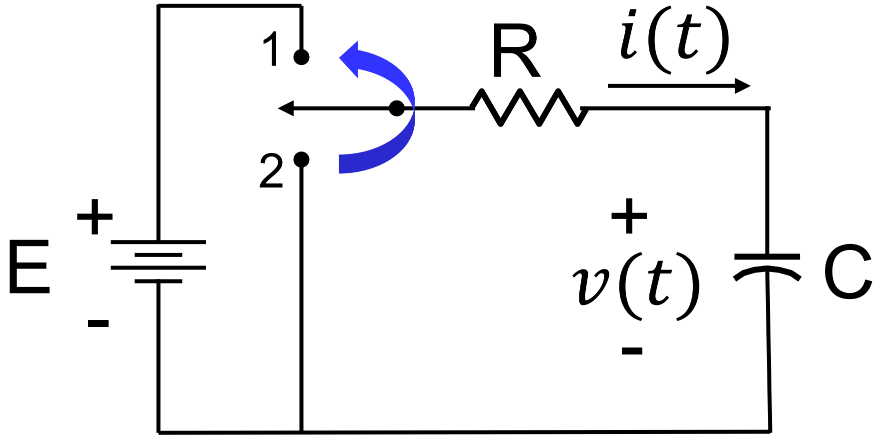
Similarly, we get:

$$\boxed{i_3 = \frac{4}{3} e^{-80t}\ \text{A} \quad t \geq 0}$$



From Example 1

What if we switch things around?



The switch was in position 2 for a very, very long time. At $t = 0$, the switch is moved to position 1.

Determine an expression for the voltage $v(t)$ for $t > 0$.

As an exercise, show that the differential equation describing our circuit for t > 0 is:

$$RC\frac{dv(t)}{dt} + v(t) = \mathrm{E}$$

The resulting equation is a **non-homogeneous differential equation**.

Solving Non-Homogeneous Differential Equations

The complete solution to the **non-homogeneous DE** is of the form:

$$x(t) = x_h(t) + x_p(t)$$

Where $x_h(t)$ is the *homogeneous* solution and $x_p(t)$ is the *particular* solution.

The *particular* solution is also called the ***forced* response**, since its form depends on the form of the forcing function (i.e. excitation source).

Solving Non-Homogeneous Differential Equations

The complete solution to the **non-homogeneous DE** is of the form:

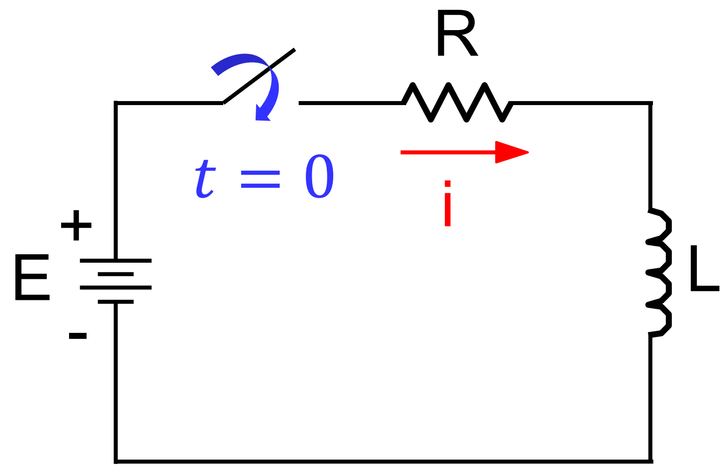
$$x(t) = x_h(t) + x_p(t)$$

For engineering applications, the components of the solution are:

1. The transient response (x_t)
2. The steady-state response (x_{ss})

$$x(t) = x_h(t) + x_p(t) \quad \Longleftrightarrow \quad x(t) = x_t(t) + x_{ss}(t)$$

Solving Non-Homogeneous Differential Equations



RL Network with DC Source

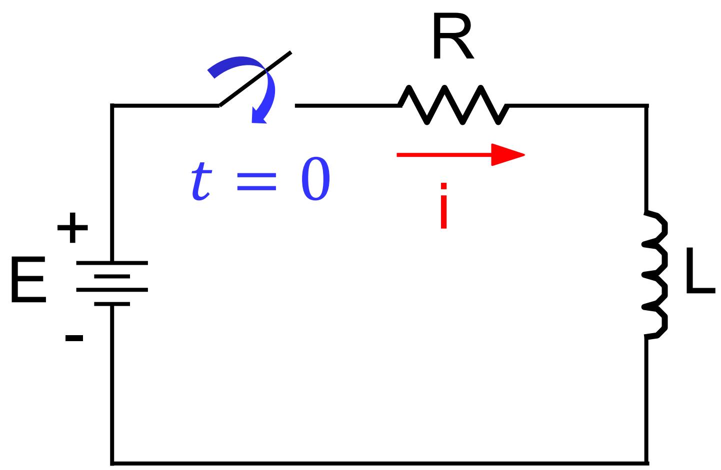
Consider the circuit shown. The switch is closed at $t = 0$. Find current $i(t)$ for $t \geq 0$.

For $t \geq 0$, we get from KVL:

$$L\frac{di}{dt} + Ri = E$$

This is a non-homogeneous differential equation.

Solving Non-Homogeneous Differential Equations



$$L\frac{di}{dt} + Ri = E$$

Transient Response:

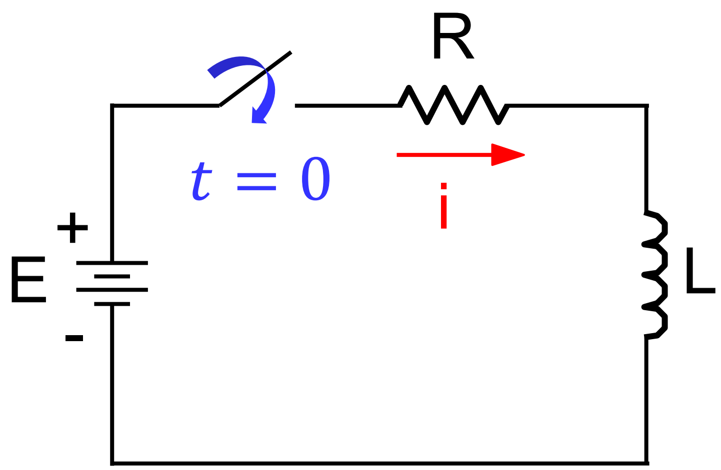
The solution of the homogeneous differential equation; that is

$$L\frac{di_t}{dt} + Ri_t = 0$$

The transient response for the RL circuit is

$$i_t = Ke^{-\frac{R}{L}t}$$

Solving Non-Homogeneous Differential Equations



$$L\frac{di}{dt} + Ri = E$$

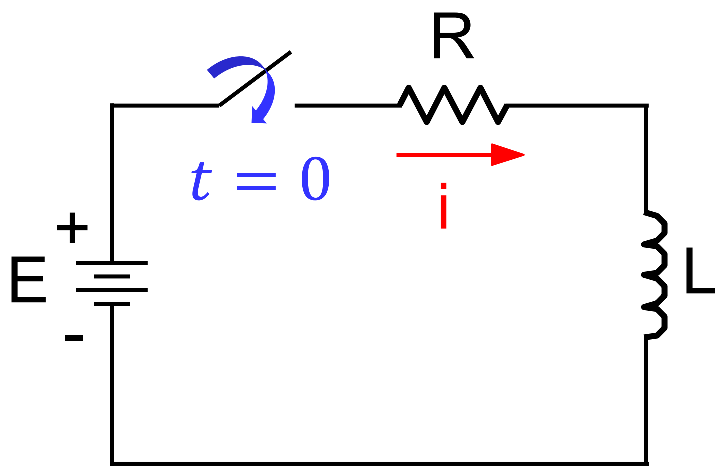
Steady-State Response:

The solution of the differential equation itself; that is

$$L\frac{di_{ss}}{dt} + Ri_{ss} = E$$

The steady-state response is similar in form to the forcing function plus all its unique derivatives.

Solving Non-Homogeneous Differential Equations



$$L\frac{di}{dt} + Ri = E$$

Steady-State Response:

For constant excitation, the steady-state response is also constant.

Let $i_{ss} = A$ (constant).

$$\frac{di_{ss}}{dt} = 0$$

Substitute in the differential equation

$$0 + RA = E$$

or

$$i_{ss} = A = \frac{E}{R}$$

Solving Non-Homogeneous Differential Equations

Complete Response: The sum of the transient response and steady-state response.

$$i(t) = i_{ss} + i_t = \frac{E}{R} + Ke^{-\frac{R}{L}t} \quad t \geq 0$$

Initial Condition: For $t < 0$, $i = 0$ since the switch is open. At $t = 0^+$, or immediately after the switch is closed, $i(0^+) = 0$ since the current in the inductor cannot change instantaneously.

Evaluate K. At $t = 0^+$, we get

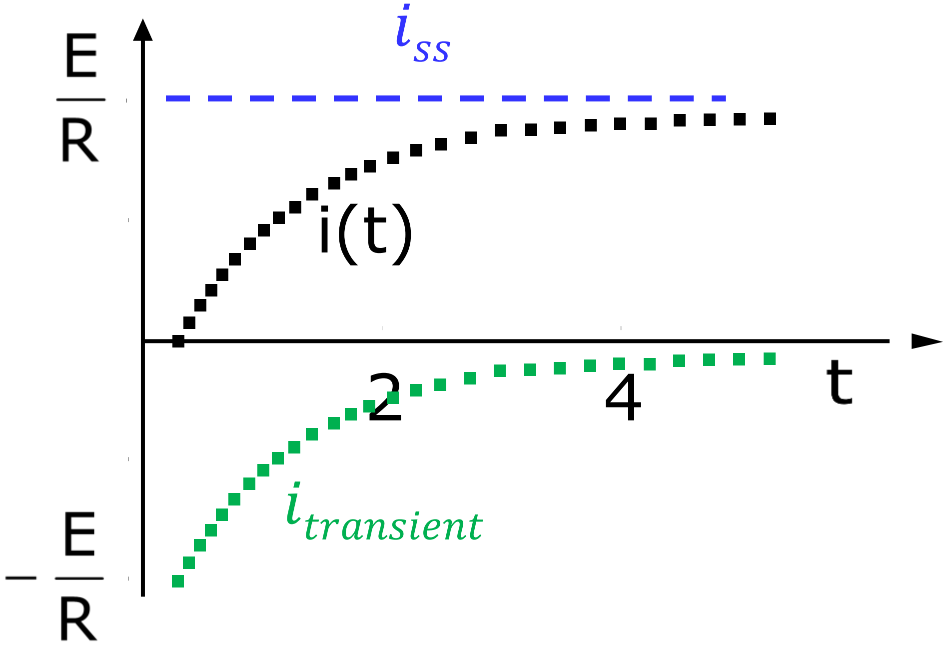
$$i(0^+) = 0 = \frac{E}{R} + Ke^0 \quad \text{or} \quad K = -\frac{E}{R}$$

Finally, we get

$$\boxed{i(t) = \frac{E}{R} - \frac{E}{R}e^{-\frac{R}{L}t} \quad t \geq 0}$$

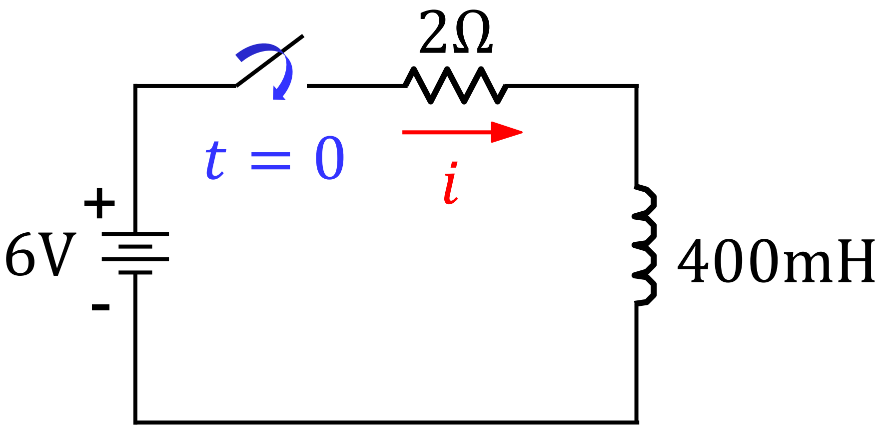
Solving Non-Homogeneous Differential Equations

A plot of the current for $t \geq 0$ is shown below.



$$i(t) = \frac{E}{R} - \frac{E}{R} e^{-\frac{R}{L}t} \quad t \geq 0$$

Example 5



In the circuit shown, the switch is closed at $t = 0$. Find current $i(t)$ for $t \geq 0$.

We derived that for an RL circuit with DC source, the response is

$$i(t) = \frac{E}{R} - \frac{E}{R} e^{-\frac{R}{L}t} \quad t \geq 0$$

Plugging in the component values,

$$i(t) = \frac{6}{2} - \frac{6}{2} e^{-\frac{2}{400m}t} \quad t \geq 0 \quad \Longrightarrow \quad \boxed{i(t) = 3 - 3e^{-5t} \quad t \geq 0}$$

Remarks

Transient Response

The transient response is the solution of the homogeneous differential equation.

(1) It is an **exponential function** whose time constant depends on the values of the electrical parameters (R, L and C);

(2) It is also called the **natural response** since it is a “trademark” of any network;

(3) It is independent of the source; and

(4) It serves as the transition from the initial steady-state to the final steady-state value.

Remarks

Steady-State Response

The steady-state response is the solution of the original differential equation.

(1) It is also called the **forced response** since its form is forced on the electrical network by the applied source;

(2) It is similar in form to the applied source plus all its unique derivatives;

(3) It is independent of the initial conditions; and

(4) It exists for as long as the source is applied.

The **forced response** is the response that will be left after the natural response dies out.

Summary

- The time constants of RL and RC circuits are $\tau = \frac{L}{R}$ and $\tau = RC$, respectively.
- For $t \geq 0$, source-free circuits result in homogeneous DEs while circuits with external sources result non-homogeneous DEs.
- The solution to a first-order homogeneous DE is an exponential function. Meanwhile, the solution to a first-order non-homogeneous DE is the sum of the transient (x_t) and the steady-state (x_{ss}) responses.